

Arduino-Based NIGHT SECURITY LIGHT

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This Arduino-based night security light can protect your home/office from thieves/intruders at night. The light glows when movement is detected in front of the PIR sensor. Simultaneously, a piezo buzzer also sounds.

Here are the main features of this night security light:

- Light turns on when it's dark and movement is detected.
- When movement is detected, the light stays on for ten seconds.
- If the bulb is already on and detects movement, it restarts the 10-second countdown.
- The bulb remains off during daylight, even when motion is detected.

The prototype of the Arduino-based flashing light is shown in Fig. 1.

Circuit and working

The circuit diagram of the Arduino-based night security light is shown in Fig. 2. It is built around an Arduino Uno board (board1), PIR sensor connected across CON2, two BC547 NPN transistors (T1, T2), a 12V one changeover relay (RL1), a small LDR (LDR1), a piezo buzzer (PZ1), a 60W 230V AC bulb connected across CON4, and few other components.

The heart of this night light is the PIR motion sensor. The PIR sensor is made of a special material that is sensitive to infrared (IR) light, which is invisible to human eyes. It can detect intrusions using passive infrared (PIR) technology. The PIR

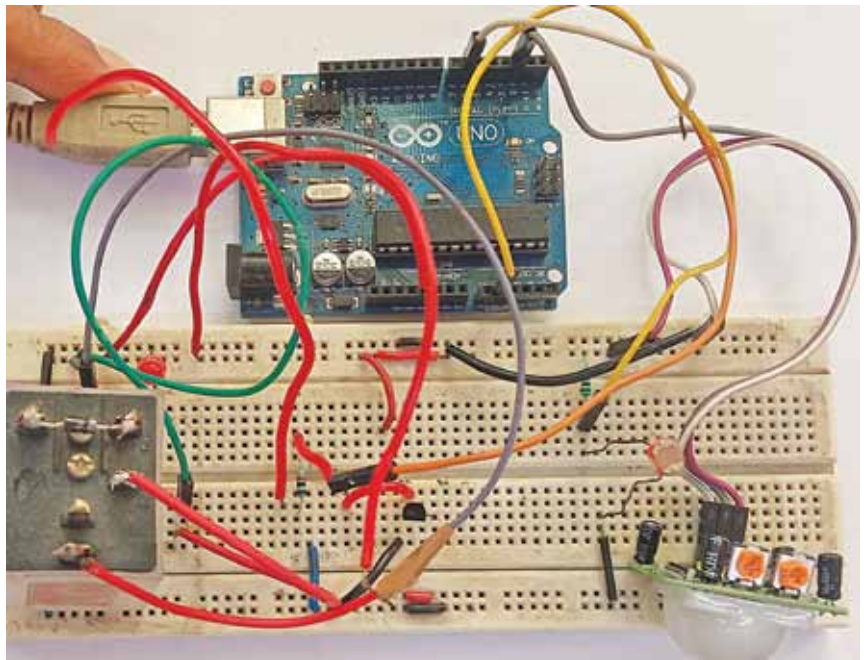


Fig. 1: The circuit tested in EFY Lab on a breadboard

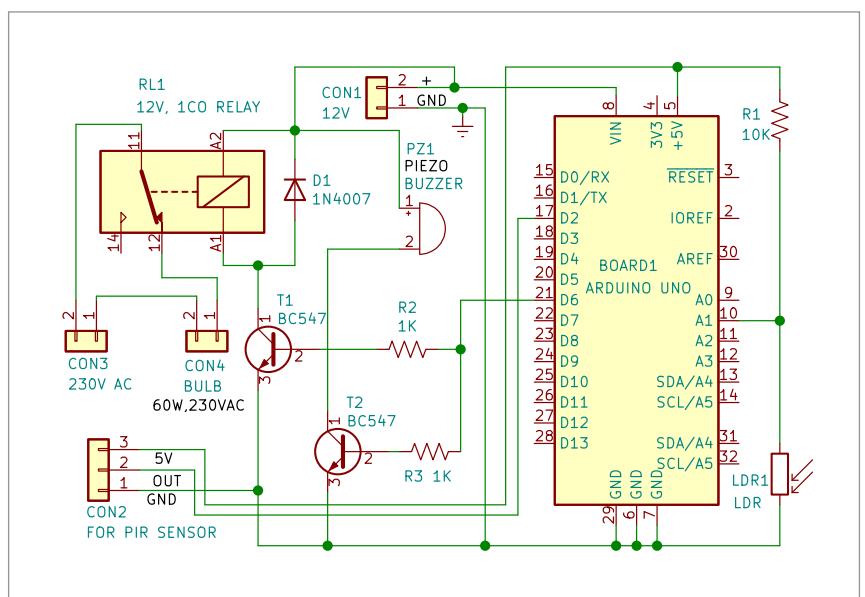


Fig. 2: Circuit diagram